

Name: Noura Maher Mohamed

Eng: Ahmed Amin

Group: ECU-DA-SU26-03

Project Title: E-Commerce Sales & Customer Analysis

Project Idea:

The project aims to analyze e-commerce sales data to understand sales performance, customer behavior, product performance, country performance, and the impact of returns and cancellations. The analysis will help identify important business trends and provide useful insights that can support business decision-making.

Dataset:

We used the **UCI Online Retail Dataset**, which contains transaction data including invoice number, product description, quantity, invoice date, unit price, customer ID, and country.

Business Questions

Sales Performance

1. What is the total revenue?
2. What is the total number of orders?
3. What is the number of orders per month?
4. How does revenue change from month to month?
5. Which month generated the highest revenue?
6. What is the average order value?
7. What is the average number of products per order?

Country Analysis

8. What are the top countries by revenue?
9. What are the top countries by number of orders?
10. What is the contribution of the top 5 countries to total revenue?
11. Which countries have the highest return and cancellation rates?

Product Analysis

12. What are the top 10 products by quantity sold?
13. What are the top 10 products by revenue?
14. Which products have the highest average unit price?

15. What is the contribution of the top 10 products to total revenue?
16. Which products have the highest number of returns/cancellations?

Customer Analysis

17. What is the number of unique customers?
18. Who are the top customers by revenue?
19. What is the average revenue per customer?

Returns & Cancellations

20. What is the return rate?
21. What is the cancellation rate?
22. What is the effect of returns and cancellations on revenue?

Data Preparation & Analysis

We used **Python** for data cleaning and analysis. The main steps included:

- Checking the dataset structure and data types.
- Handling missing values.
- Removing duplicate records.
- Cleaning product descriptions.
- Creating a new **Revenue** column using:
Revenue = Quantity × UnitPrice
- Identifying returns and cancellations.
- Analyzing sales by month, country, product, and customer.
- Creating visualizations to represent the main findings.

Visualization & Dashboard

We used **Matplotlib** to create visualizations for the analysis, including monthly revenue, monthly orders, top countries, top products, and top customers.

We also created a **Power BI Dashboard** containing the main KPIs and visualizations to present the results in an interactive and easy-to-understand way.

Tools Used

- Python
- Pandas

- NumPy
- Matplotlib
- Excel
- Power BI

والحاجات اللي مش React واضحة، أكاديمية، ومش فيها تفاصيل زيادة عن الموقع والـ: كده مناسبة جداً تتبعت للمعيد أساسية للمشروع.